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Emotional Speech-driven 3D Body Animation via Disentangled Latent Diffusion

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Abstract

Existing methods for synthesizing 3D human gestures from speech have shown promising results, but they do not explicitly model the impact of emotions on the generated gestures. Instead, these methods directly output animations from speech without control over the expressed emotion. To address this limitation, we present AMUSE, an emotional speech-driven body animation model based on latent diffusion. Our observation is that content (i.e., gestures related to speech rhythm and word utterances), emotion, and personal style are separable. To account for this, AMUSE maps the driving audio to three disentangled latent vectors: one for content, one for emotion, and one for personal style. A latent diffusion model, trained to generate gesture motion sequences, is then conditioned on these latent vectors. Once trained, AMUSE synthesizes 3D human gestures directly from speech with control over the expressed emotions and style by combining the content from the driving speech with the emotion and style of another speech sequence. Randomly sampling the noise of the diffusion model further generates variations of the gesture with the same emotional expressivity. Qualitative, quantitative, and perceptual evaluations demonstrate that AMUSE outputs realistic gesture sequences. Compared to the state of the art, the generated gestures are better synchronized with the speech content, and better represent the emotion expressed by the input speech. Our code is available at amuse.is.tue.mpg.de.

1. Introduction

Animating 3D bodies from speech has a wide range of applications, such as telepresence in AR/VR, avatar animation in games and movies, and to embody interactive digital assistants. While methods for speech-driven 3D body animation have recently shown great progress [5, 7, 31, 56, 101], existing methods do not adequately address one crucial factor: the impact of emotion from the driving speech signal on the generated gestures. Emotions and their expressions play a fundamental role in human communication [29, 35, 65] and have become an important consideration when designing computer systems that interact with humans in a natural manner [78, 79]. They are of central concern when synthesizing human animations for a wide variety of application contexts, such as Socially Interactive Agents [61]. Because of this, speech-driven animation systems must not only align movement with the rhythm of the speech, but should also be capable of generating gestures that are perceived as expressing the suitable emotion.

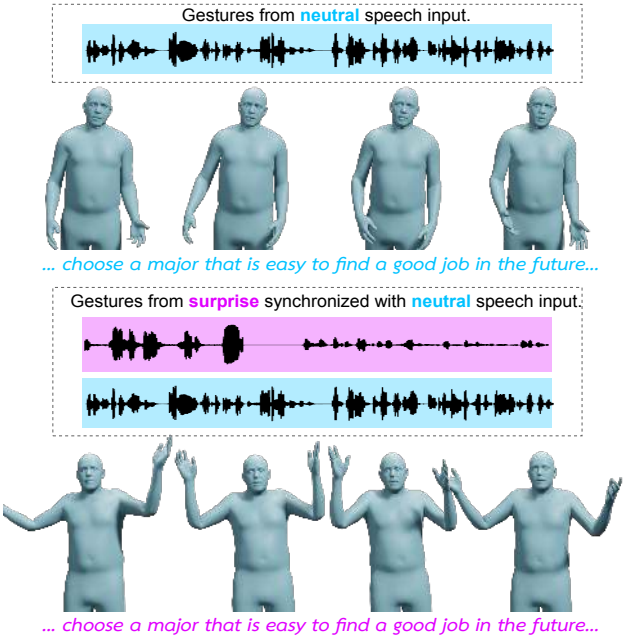


Figure 1. **Goal.** AMUSE generates realistic emotional 3D body gestures directly from a speech sequence (top). It provides user control over the generated emotion by combining the driving speech sequence with a different emotional audio (bottom).

Many factors contribute to the perception of emotion and personal idiosyncrasies, such as facial expressions [19], gaze and eye contact [42], physiological responses [47], tone of voice [87], body language [66], and gestures [39]. When it comes to 3D animation, the most relevant factors

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are facial expressions, gestures, and body language [95]. While emotional speech-driven animation methods have recently been proposed for 3D faces [18, 74, 90, 107], animating emotional bodies from speech remains under-explored.

Generating gestures solely from speech with emotional control is a difficult task. First, the mapping from audio to body motion is a non-deterministic many-to-many mapping, which is difficult to model. Gestures across subjects can vary when uttering the same sentence, and a single individual's motions can change significantly across repetitions. Second, factoring out the impact of emotional state on the body motion from other, unknown factors, is difficult. This requires disentangling the effects of three different factors on the generated motion, namely content-based (i.e., gestures related to speech rhythm and word utterances), emotion-based, and those based on personal style. AMUSE addresses this by separating a speech sequence into content, emotion, and style latent vectors, which are then used to condition a latent diffusion model. Specifically, AMUSE consists of three main components: (1) an audio autoencoder trained to produce disentangled vectors of content, emotion, and style, (2) a 3D body motion prior in the form of a temporal variational autoencoder (VAE) to generate smooth and realistic gestures, and (3) a latent diffusion model, which generates 3D body motion given the input content, emotion, and style latent vectors.

Training such a model requires a speech-to-3D body dataset of sufficient scale, which is rich and diverse in speakers and emotions. BEAT [55] is a good candidate because it provides a large set of 3D gestures associated with single-person monologues. Unfortunately, the bodies are represented as skeletons, and it lacks face mocap markers and FLAME expressions. Instead, to produce realistic body animations, we require articulated 3D body surfaces. To overcome this, we convert BEAT sequences to SMPL-X [73] format using MoSh++ [62] and use the SMPL-X parameters for training. See [56] for comparison.

Our contributions are: (1) We present a framework to synthesize emotional 3D body gestures directly from speech. (2) We factor an input audio into disentangled content, emotion and style vectors, which enables us to separately control emotion in generated gestures. (3) We adapt temporal latent diffusion for multiple target conditions.

2. Related Work

2.1. 3D Conditional Human Motion Generation

Early works focus mostly on predicting [10, 16, 33, 41, 57, 64, 70, 86, 106, 109] or generating human motion [30, 49], but do not consider multi-modal control. Recently, conditional motion generation through other modalities, such as text [2, 8, 9, 17, 22, 28, 77], music [50, 68, 94], speech [32], or action labels [27, 75], has gained more attention. Below, we focus on speech-driven motion generation methods, since they are the most relevant to our work.

2.2. Gesture Generation from Speech

Rule-based gesture synthesis. Embodied conversational agents (ECA) are designed to interact and communicate with humans. Using the Behavior Markup Language (BML) [44] one can build rule-based systems for humanoids based on predefined behaviors [80]. This is used for completion of a storytelling task in an expressive manner [45]. The BEAT rule-based toolkit [14] enables adding non-verbal behavior on top of a pre-animated figure. Thiebaux et al. [92] develop an ECA by using procedural animation techniques and keyframe interpolation. Marsella et al. [63] design a generalized rule-based agent to generate expressions, eye gaze, and gestures from speech. Each of these approaches are based on non-trainable, rule-based techniques that may require substantial manual modelling effort to adapt to new tasks.

Data-driven gesture synthesis. More recently, data-driven methods have superseded rule-based systems. Yoon et al. [104] use a fusion of text, audio and upper body gestures to learn an upper body gesture avatar, but can only control the style of individual speakers by sampling from their latent space. SpeechGestureMatching [32] generates 3D facial meshes and 3D keypoints of the body and hands from speech, but the outputs are separated and the method does not provide control over the generations. QPGesture [98] uses phase to better align the generated 3D skeleton-based gesturing avatars with the audio input. Ginosar et al. [23] and Diverse-3D-Hand-Gesture-Prediction [82] generate hand and arm motions only. Audio2Gestures [48] encode motion and audio to a low-dimensional latent space and generate gestures. SEEG [53] aims to generate gestures that align well with the semantics of the speech. Diff-TTSG [67] regresses speech and gestures at the same time, joining the two modalities in a single system. DiffGAN [3] retargets gestures across speakers in a low-resource setting. The GENEA challenge [105] tackles gesticulation from speech alone using the Talking-with-Hands dataset [46]. Gesture2Vec [99] uses a machine translation model to translate text into gesture chunks and output full sequences using such quantized representations. TalkSHOW [101] uses a VQ-VAE to generate 3D human bodies gesturing with facial expressions from speech segments, but in an uncontrolled manner. Similarly, Co-speech gesture [60] uses an RQ-VAE to generate different gestures from speech. Alternative gesture generation from speech methods have been proposed such as reinforcement learning [91], self-supervised pre-training [40], and diffusion [67, 110]. BodyFormer [71] introduces a dataset of pseudo-groundtruth and a transformer-based method for generating gestures from speech. However, none of these methods provide explicit emotional con-

trol over the generated motion.

For controllable generation, GestureDiffuCLIP [7] incorporates multiple conditions including CLIP [83] text features, video, or motion prompts via AdaIn [37] layers to generate gestures from speech, however, it does not allow explicit control over the emotion conveyed by the driving audio. ListenDenoiseAction [5] combines conformers and the DiffWave [43] architecture to generate gestures that can be controlled by a style vector, RhythmicGesticulator [6] disentangles the latent space into a vector related to the semantics of the gesture and one related to the subtle variations, while DisCo [54] models content and rhythm. StyleGestures [4] adapts MoGlow [34], demonstrating limited control over some motion attributes like the speed and expressiveness of gestures. DiffuseStyleGesture [97] uses diffusion to generate diverse gestures from speech.

2.3. Emotion Control

Emotion classification and control has been little studied in 3D human motion generation with only a a few methods using skeletal motion in multi-class classification. Ghaleb et al. [20] employ a spatio-temporal graph convolution network to classify gestures into four classes: preparation, stroke, retraction, and neutral. Li et al. [52], on the other hand, use hidden Markov models for emotion classification of human movement mocap data. Karras et al. [38] learn face animations of a single actor, and test their method on different tasks by modifying the latent vectors. However, there is no disentanglement mechanism, and they do not model the synchronization of the emotion with the with the facial motions. Recently, EmoTalk [74], animates emotional 3D faces from speech input with control over the emotion intensity and EMOTE [18] disentangles emotion and speech to allow emotion editing at test time. However, models solely intended for facial tasks like lip syncing and capturing expressions might not smoothly adapt to the complexity of whole-body movements and distinct articulation. Regarding emotion-conditioned motion generation, Aberman et al. [1] show style-transfer from video data to motion and provide some style-based control, but do not address speech-driven emotional gestures. Similarly, the ZeroEGGs [21] dataset contains some emotional gesture controls but also includes more generic styles of motion. The method requires the input of arbitrary frames of desired motion to encode a style, thereby relying on motions and speech as conditions during inference. Text-driven emotional gesticulation, as explored by Bhattacharya et al. [11, 12], emphasizes the generation of gestures based on textual cues, incorporating additional conditions such as speech, speaker ID, seed poses, as well as valence, arousal, and dominance triplets. However, these approaches do not provide the means to distill explicit emotion features, limiting free control over the generated gestures. Closer to our work, EMoG [102] incorporates emotion cues from the BEAT dataset [55] to generate improved gesture quality without explicit emotion control. EmotionGesture [81] uses a TED Emotion Dataset and BEAT to incorporate emotion features in gesture generation and generate emotional gestures. Although they can generate emotional gestures, their method is not end-to-end and has no explicit motion control. Specifically, it uses an emotion-conditioned VAE after training to acquire diverse emotion features that are used to generate gestures without guarantees and control over emotion types. Wu et al. [96] introduce the first multi-cultural gesture dataset containing 200 individuals of 10 different cultures. In contrast to prior work, we explicitly control the emotions conveyed by the generated gestures solely through emotional speech without relying on additional conditions.

3. Method

The AMUSE pipeline consists of two separately trained networks. The audio disentanglement module, which encodes input speech into latent vectors for content, emotion, and style is described in Sec. 3.2. The main architecture is described in Sec. 3.3. It consists of a 3D human motion prior coupled with a latent diffusion model. It takes random noise (or partially denoised latent vectors) on the input and outputs a human motion sequence. We introduce broader applications in gesture editing in Sec. 3.4.

3.1. Preliminary: Expressive 3D Body Model

SMPL-X [73] is a 3D model of the body surface. SMPL-X is defined as function $M(\boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\psi})$ that produces a 3D body mesh. It is parameterized by identity shape $\boldsymbol{\beta} \in \mathbb{R}^{300}$, pose $\boldsymbol{\theta} \in \mathbb{R}^{J\times 3}$ including finger articulation for rotations around J joints, and facial expression $\boldsymbol{\psi} \in \mathbb{R}^{100}$. We adopt the continous 6D rotation representation for training following Zhou et al. [108], making $\boldsymbol{\theta} \in \mathbb{R}^{J\times 6}$. Given pose parameters and any shape parameter, we can obtain body mesh vertices V using the differentiable SMPL-X layer [73]. As the focus of our paper is on synthesizing body gestures and not locomotion, we disregard 8 joints that correspond those of the lower body joint poses, leaving $J = 47$. Further, we omit the facial expression parameters, i.e., set $\boldsymbol{\psi} = \mathbf{0}$.

3.2. Speech Disentanglement Model

Architecture. The goal of the this model is to factor an input speech into three disentangled latent representations, one for content (i.e., the words spoken), one for emotion, and one for personal style. To do so, we devise a specialized encoder–decoder architecture with three separate encoders, one for each latent space. We denote the encoders as: $E_c(a) = c, \quad E_e(a) = e, \quad E_s(a) = s$, where a is the input filterbank, c, e and s denote the latent vectors for content, emotion and style and E_c, E_e and E_s are their encoders. The architecture of the three encoders follows the

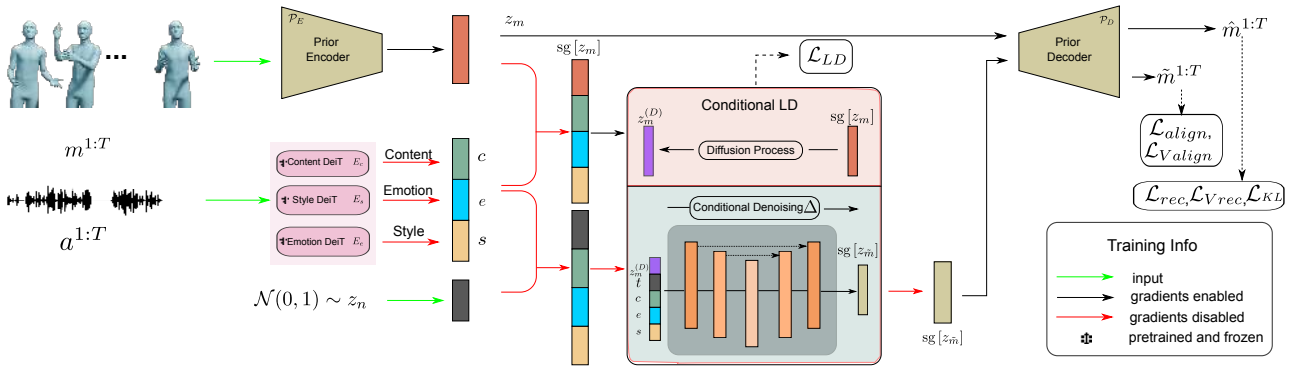


Figure 2. **Training.** We train the motion prior ($\mathcal{P}_E, \mathcal{P}_D$) and the latent denoiser Δ jointly, while keeping the audio encoding networks frozen. In the forward pass, we take an input audio $a^{1:T}$ and pose sequence $m^{1:T}$. Firstly, we do a forward pass of $m^{1:T}$ through $\mathcal{P}_E$ and $\mathcal{P}_D$ and compute $\mathcal{L}_{rec}$, $\mathcal{L}_{Vrec}$, and $\mathcal{L}_{KL}$. Then, we apply the diffusion process to a gradient-detached sg $[z_m]$ obtaining the noisy $z_m^{(D)}$, which is then denoised with Δ and $\mathcal{L}_{LD}$ is computed. Finally, we use Δ to fully denoise z_n into gradient-detached sg $[z_{\tilde{m}}]$, further decode $\tilde{m}^{1:T}$ using $\mathcal{P}_D$, and compute $\mathcal{L}_{align}$ and $\mathcal{L}_{Valign}$.

design by Gong et al. [25, 26] (i.e., leveraging the DeiT visual transformer [93] adapted for processing filterbank images extracted from the input audio). The decoder takes the three latent vectors and produces a reconstructed filterbank. Formally $D(c, e, s) = \widehat{a}$, where $\widehat{a}$ denotes the reconstructed filterbank. The decoder architecture consists of a fusion module and transformer-encoder layers.

Training. The audio module is trained with a multiple loss terms that ensure that the three latent spaces are properly disentangled. In addition to the standard autoencoder reconstruction loss, we also employ three cross-reconstruction losses, in which we enforce the correct reconstruction of the audio signal where we modify one of the content, style or emotion latents. Additionally, we employ three loss terms on the latent vector predictions – namely emotion and style classification losses over e and s, and a content similarity loss between pairs of two content latent vectors extracted from audios that have the same spoken content. For a detailed description of the encoder–decoder architecture, a formal definition of the loss functions and a detailed description of the training process please refer to the Sup. Mat.

3.3. Gesture Generation Model

Motion prior. Similar to [15, 76], our motion prior network is a VAE transformer architecture with encoder $\mathcal{P}_E$ and decoder $\mathcal{P}_D$. Specifically, both $\mathcal{P}_E$ and $\mathcal{P}_D$ follow a U-Net-like [85] structure with skip connections between transformer blocks (see Sup. Mat. for details). The positional embeddings are learnable and injected into each multi-head attention layer, following the design of Carion et al. [13]. Formally, the encoder takes a sequence of T frames of the SMPL-X pose vectors $m^{1:T} \in \mathbb{R}^{6J \times T}$ and the first two tokens of its output, $\mu \in \mathbb{R}^{d_m}$ and $\Sigma \in \mathbb{R}^{d_m \times d_m}$ are used to extract the motion latent $z_m \in \mathbb{R}^{d_m}$ via the reparametrization trick. The decoder takes zero positional encodings as query input and the motion latent is fed as memory to every cross-attention transformer layer, producing the reconstructed motion $\hat{m}^{1:T}$.

Diffusion process. The forward diffusion process is similar to [36, 69]. We employ fixed variance and linearly scaled noise scheduler. We add noise to the motion latent z_m for D diffusion timesteps to obtain $z^{(D)}$ following:

$$q(z_m^{(t_d)} \mid z_m^{(0)}) = \mathcal{N}(z_m^{(t_d)}; \sqrt{\bar{\alpha}_{t_d}} z_m^{(0)}, (1 - \bar{\alpha}_{t_d})\mathbf{I}),$$

with $\alpha_{t_d} = 1 - \beta_{t_d}$, $\bar{\alpha}_{t_d} = \prod_{s=1}^{t_d} \alpha_s$, and β_{t_d} denotes diffusion process variance.

Conditional denoising process. The denoising process consists of iteratively denoising a conditioned noisy motion latent vector to obtain the denoised motion latent $z_{\tilde{m}^{1:T}}$. Our denoiser Δ is a latent variable model [84] and its architecture is similar to the U-Net-like structure of the motion prior encoder $\mathcal{P}_E$. The input of the model is a concatenation of: $z_m^{(t_d)}, \text{SE}(t_d), c, e, s \in \mathbb{R}^{256}$, where $\text{SE}(t_d)$ is a sinusoidal positional encoding of diffusion timestep t_d as defined in [36]. Δ iteratively denoises through each reversed diffusion step:

$$z_m^{(t_d - 1)} = \Delta([z_m^{(t_d)}, \text{SE}(t_d), c, e, s]).$$

Training. We optimize the motion prior and the latent denoiser jointly to ensure audio–motion latent code alignment during conditional fusion in the denoising process using a 3-step forward pass through the gesture generation model. First, following standard VAE practice, we reconstruct $\hat{m}^{1:T}$ by the motion prior forward pass. As shown in Fig. 2, we then disable gradient calculation in $\mathcal{P}_E$ to infer the intermediate motion latent sg $[z_m]$, which serves as input to the denoiser. At this stage, we obtain the denoiser noise prediction, δ and use to compute the diffusion model gradients. Finally, in the third step we com-

pute $\tilde{m}^{1:T} = \mathcal{P}_D(\text{sg}[z_{\tilde{m}}])$, where $z_{\tilde{m}}$ is obtained by iteratively using the Δ to obtain a fully denoised latent from $z_n^{(t_D)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. We indicate computations done without gradients with a stop-gradient operation sg [.].

Losses. To train the motion prior, we include the standard VAE losses, namely the reconstruction loss on pose parameters $\mathcal{L}_{rec}$ and on vertex coordinates $\mathcal{L}_{Vrec}$ using the smooth L1 metric introduced in [24], which we denote as L_1^s:

$$\mathcal{L}_{rec} = L_1^s(m^{1:T}, \hat{m}^{1:T}), \quad \mathcal{L}_{Vrec} = L_1^s(V^{1:T}, \hat{V}^{1:T}),$$

where the root-centered vertices V are obtained by feeding in pose parameters m to a differentiable SMPL-X layer (without learnable parameters) and a mean shape $\boldsymbol{\beta} = \vec{0}$. The KL divergence loss of the motion prior is:

$$\mathcal{L}_{KL} = \frac{1}{2}\left[\sum_{i=1}^{z}(\mu_i^2 + \sigma_i^2) - \sum_{i=1}^{z}\left(log(\sigma_i^2) + 1\right)\right].$$

To ensure the alignment of the diffusion-generated motions and the input audio, we apply the alignment reconstruction loss on the inferred motion pose parameters and the vertex coordinates:

$$\mathcal{L}_{align} = L_1^s(m^{1:T}, \tilde{m}^{1:T}), \quad \mathcal{L}_{Valign} = L_1^s(V^{1:T}, \tilde{V}^{1:T}).$$

Finally, we utilize the objective similar to [15, 36, 84] to supervise the denoiser:

$$\mathcal{L}_{LD} = \left\| \delta^{(t_d)} - \Delta(z_m^{(t_d)}, \text{SE}(t_d), c, e, s) \right\|_2^2,$$

where $\delta^{(t_d)}$ is the noise vector sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ in the corresponding diffusion step t_d. The combined gesture model loss is:

$$\mathcal{L}_{ges} = \mathcal{L}_{rec} + \mathcal{L}_{Vrec} + \mathcal{L}_{KL} + \mathcal{L}_{align} + \mathcal{L}_{Valign} + \mathcal{L}_{LD}$$

Inference. We employ DDIM [89] to infer high quality conditional motion samples with a small number of denoising timesteps. During inference we draw a sample vector from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ to iteratively denoise in reversed timesteps. The denoised sample is then passed through the decoder $\mathcal{P}_D(z_{\tilde{m}^{1:T}})$ to obtain motion $\tilde{m}^{1:T}$.

3.4. Gesture Editing

Due to the disentangling of the inputs, AMUSE achieves semantic gesticulation control using two driving input audios. Specifically, given two input audio signals a_1 and a_2, we extract their latent representations of content c_1, c_2, emotion e_1, e_2, and style s_1, s_2. Then, we simply initialize the denoising procedure of Δ with the triplet (c_1, e_2, s_1), generating the gesture with the content and style of a_1 but the emotion of input audio a_2. Similarly, instead of emotion we can also change the gesticulation style to that of the speaker of a_2 by initializing with (c_1, e_1, s_2).

4. Implementation Details

MoCap data preparation. The BEAT [55] mocap sequences, captured in a Vicon system at 120 Hz, are downsampled to 30 Hz and processed using MoSh++ [58, 62] to obtain SMPL-X parameters. Given a sequence of 3D mocap marker positions, we jointly optimize SMPL-X shape and pose parameters, 3D body translation, and embedding of the mocap markers in the SMPL-X surface. Once processed, the sequences are then divided according to the emotion annotations in the BEAT dataset. We use sequences of English speaking subjects in monologue speaking style for training and evaluating AMUSE. For each sequence we draw $m^{1:L}$ at 30 FPS and concatenate with audio content c, emotion e, and style s latent vectors. Then, we segment it to 10-sec windows T, beginning from the timestamp 0 and discarding additional unaligned information at the end. This preprocessing choice allows us to train transformer networks without masking. We provide additional data processing information in the Sup. Mat.

Audio preprocessing. We use audio sequences belonging to eight categorical emotion labels (neutral, happy, angry, sad, contempt, surprise, fear, and disgust). Each audio chunk of 10s is converted into a filter bank with 128 mel-frequency bins with a 25ms Hamming frame window and 10ms frame shift. We mask each sample with a maximum length of 24 in the frequency domain and a maximum length of 96 in the time domain, employing Park et al. [72]. Following [25, 26], we standardize the filter bank and augment it via noise injection and circular shifting. Before feeding in our speech disentanglement model, each filter bank is split into a sequence of fixed 1209 patches of 16 x 16 each having 6 units overlap in frequency and time domain.

Motion prior. The motion prior is a VAE encoder–decoder with 9 layers and 4 heads, following Chen et al. [15]. The encoder–decoder is a U-Net-like transformer with residual connections. Learnable positional embeddings are injected in each multi-head attention layer. We have a linear projection at the start and the end of our motion prior network. The KL divergence term is weighted with a factor of $1e-4$.

Denoiser. The denoiser follows the same network architecture as our prior encoder. The hidden dimension of all transformer layers is 1024. We use 1000 diffusion steps D during training and 50 during inference. Noise betas are in range $[0.00085, 0.012]$. We jointly optimize the prior and denoiser networks for 5000 epochs with batch size of 64, learning rate 0.0001, and the AdamW optimizer [59].

5. Experiments

Speech disentanglement model. We evaluate the performance of the speech disentanglement model quantitatively using classification accuracy and F1 scores on emotion and style. The accuracy is computed as average scores

for all 8 emotion as well style categories that are part of the test dataset. The emotion and style accuracy is 91.53% and 96.06%, respectively. The emotion F1 score and style F1 scores are 0.914 and 0.960, respectively. See the Sup. Mat. for ablations and a detailed metric analysis.

Gesture generation model. We evaluate the performance of our gesture generation model quantitatively, qualitatively, and perceptually against following methods: TalkSHOW [101] and the re-implementation of Habibie et al. [31] provided by the TalkSHOW authors in the official TalkSHOW release [100], DiffuseStyleGesture (DSG) [97], MoGlow [34], and CaMN [55]. Additionally, we adapt TalkSHOW to include categorical emotion labels as input along with the existing architecture that only allows one-hot encodings of personal style. We then retrain it on our training data. We refer to it as TalkSHOW-BEAT. There are some concurrent works [5, 7, 56], which introduce methods for gesture generation from speech, however, direct comparison is hindered by the unavailability of released code our task. Refer to the Sup. Mat. for the ablation experiments, the emotion and style editing experiments, and their quantitative evaluation.

5.1. Quantitative Evaluation

To quantitatively evaluate our method's gesture generations and edited gesture generations, we train a transformer-based encoder architecture (denoted as M) similar to Petrovich et al. [76] in an autoencoder setting, where we append a CLS token at the beginning of the motion sequence. M is trained with a cross-entropy emotion classification objective applied to the output CLS token. We train M on the BEAT training dataset and use its features to compute the following metrics: (1) Fréchet gesture distance (FGD): We follow [88, 103, 104] to compute the feature distance between generated and ground truth motion features. (2) Gesture diversity (Div): Similarly to Chen et al. [15], we compute variance across generated features. (3) Gesture emotion accuracy (GA): We report top-1 emotion classification accuracy predicted by a classifier trained on the motion M-predicted latents. (4) Beat align (BA): We follow [51, 55], to evaluate the motion-speech correlation in terms of the similarity between the kinematic motion beats and speech audio beats. The kinematic motion beats are directly computed from the generated motion sequences. (5) Semantic-Relevant Gesture Recall (SRGR): We follow Liu et al. [55], to evaluate the semantic relevancy of gestures with GT motion. We use the ground truth semantic scores to compute this metric. The scores are obtained from the BEAT authors, representing a continuous score on a scale 0-1 per gesture style for 4 gesture semantic categories: beat, deictic, iconic, and metaphoric. While comparing with methods that output coarse skeletal data (DSG [97], MoGlow [34], and CaMN [55]), we convert the skeleton motion data into the SMPL-X axis angle representation. For details on the architectures and training of M, and the losses, please refer to the Sup. Mat.

Method	SRGR↑	BA↑	FGD↓	Div→	GA[a]↑
GT	—	0.83	—	27.83	64.04
Ours	0.36	0.81	388.63	25.06	46.76
Ours-EmoEdit[b]	—	0.79	792.58	24.68	34.18
TalkSHOW-BEAT	0.31	0.64	808.99	24.16	22.71
TalkSHOW [101]	0.30	0.60	762.15	23.19	29.41
DSG [97]	0.23	0.40	763.10	19.77	22.70
Habibie et al. [31]	0.23	0.39	809.17	21.34	16.67
MoGlow [34]	0.21	0.35	1097.03	19.50	16.62
CaMN [55]	0.21	0.39	1063.87	18.90	14.17

[a] GA is average of all 8 emotions.
[b] GA for these are average accuracy for all generations with 7 edited audio sequences.

Table 1. **Gesture quantitative results.** We compare our methods against several SOTA methods using metrics explained in Sec. 5.1. We observe that AMUSE outperforms in all scores compared to baseline methods. Additionally, AMUSE-EmoEdit outperforms in Beat Align, Diversity, and Gesture Emotion Accuracy scores compared to the baseline methods.

We prepare the evaluation data by randomly selecting 72 unique motion sequences each of length 10s and comprising 8 emotions across test subjects and compute the aforementioned metrics. We use 9 sequences for each emotion per subject. The results are reported in Tab. 1. All best scores are highlighted in green and second best in blue. AMUSE outperforms the baseline methods in all given metrics. To validate the performance of gesture emotion editing, we also report the same metrics for the emotion editing task (Ours-EmoEdit). During inference, the input style and content latents are extracted from neutral-emotion audio, while the emotion latent comes from a different audio of different emotion. These emotional edits offer numerous possibilities, allowing for transitions from any to any emotion. Tab. 1 shows the average for editing from neutral to other emotions. Since we require the GT gesture semantics score to compute SRGR metric, it is not possible to compute the SRGR for the synthetic edited-emotion gestures as they are not part of the original BEAT dataset. Ours-EmoEdit outperforms the baseline methods in BA, Div, and GA metrics. This demonstrates the capability of our model to maintain highly discriminative cues when switching between different emotions. TalkSHOW-BEAT has the second best score for SRGR whereas TalkSHOW demonstrates second best FGD score. Although, our model and ours-EmoEdit show improvements over the baseline methods, GT motions have higher diversity, Beat alignment score, and are easier to classify than generations of AMUSE, highlighting the challenging nature of the problem.

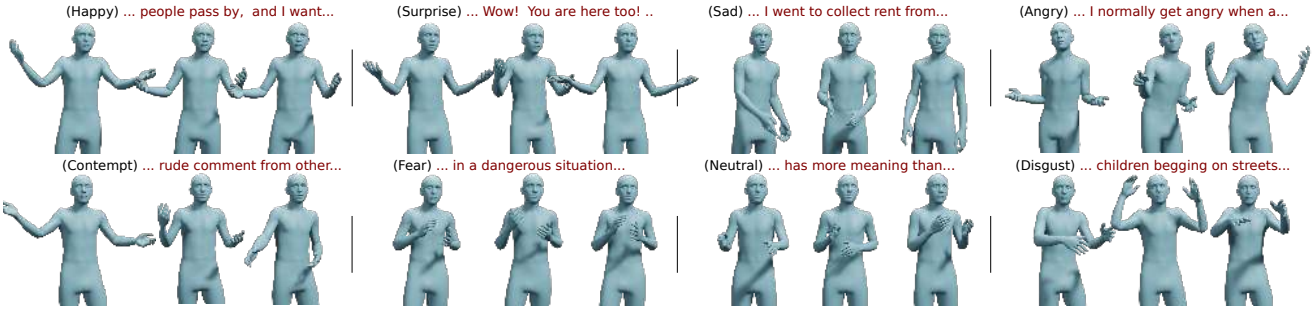


Figure 3. **Qualitative comparison across all emotions.** We evaluate generation on different test audios. AMUSE exhibits well-synchronized beat gestures and consistently produces gestures that accurately convey the emotional content expressed in the input speech.

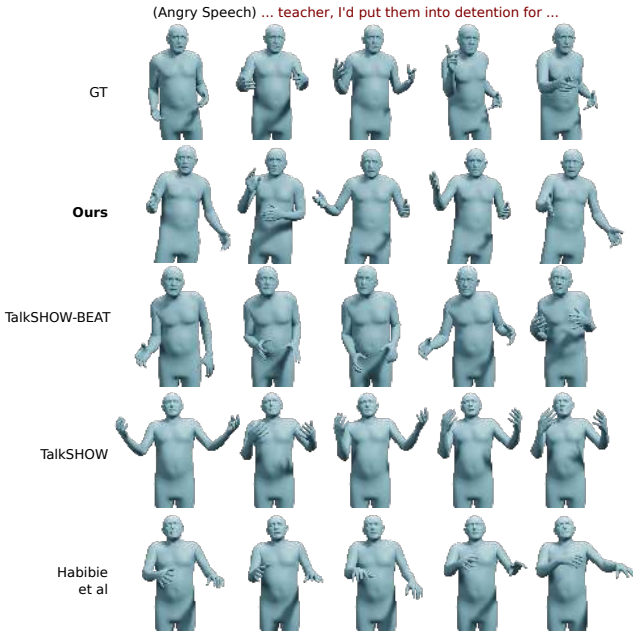


Figure 4. **Qualitative comparison with baseline methods**. The speech segment describes intense angry speech.



Figure 5. **Qualitative evaluation of diverse generations**. Multiple generations overlayed.

5.2. Qualitative Evaluation

Comparison with baseline methods. In Fig. 4, we demonstrate comparison with baseline methods that output a 3D body mesh: Habibie et al. [31], TalkSHOW [101], TalkSHOW-BEAT, and the BEAT ground truth (GT) [55]. We observe that AMUSE generates gestures that are semantically closer to the speech content and produces expressive emotional gestures closer to the perceived emotion. For example, the GT motion exhibits anger when saying "*put them into detention*". AMUSE demonstrates tense posture and aggressive movements comparable with the ground truth data and accurate synchronization with the spoken words. TalkSHOW [101] and Habibie et al. [31] exhibit limited movement and display inferior and static gestures on test audios as seen in the last two rows of Fig. 4. TalkSHOW-BEAT slightly outperforms other baseline methods by demonstrating enhanced synchronized gestures, but it still does not perform as well as AMUSE.

Diverse emotional gestures. In Fig. 5, our probabilistic model can generate diverse gestures for same input audio.

Emotional gesture generation. In Fig. 3 AMUSE demonstrates strong correlation with the spoken utterances as well as different emotions. We observe that our model is able to correlate semantic words to associated gestures. For example, gestures demonstrate forceful actions and tense stance with angry audio "*normally get angry*" whereas it generates lowered and calm hand positions for sad audio "*I went to collect*". Similarly, our generations show hands that are closer to body for fearful audio "*in a dangerous situation*" while widely open expressing astonishment for happy and surprised audio "*people pass by*" and "*Wow! You are here*".

Emotion editing. We use two audio streams of a female subject for neutral and sad emotion. This experiment edits the subject's gesture style from moderately controlled hand movements to a sad style with lethargic posture conveying a sense of heaviness, as seen in Fig. 7 (top).

Gesture style editing. We use audio streams of two male subjects for the happy (ID - 13) and angry (ID - 2) emotion. With the emotion, style and content latent fusion mechanism from two driving audio streams, AMUSE is able to adapt the male (ID - 13) subject's body gestures from being close to their body to more open with squared tightened shoulders, expressing a shift from happy to angry emotions of a different subject (ID - 2), as shown in Fig. 7 (bottom). Please refer to the supplemental video for qualitative results and comparisons to additional gesture genera-

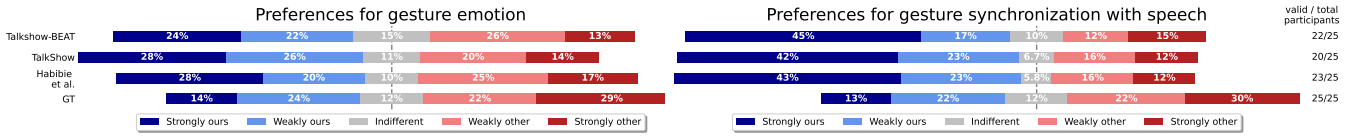


Figure 6. **The perceptual study results for gesture emotion preference (left) and synchronization with speech (right)**. The number of attentive participants that passed the catch trials is indicated on the right and the reported results only consider these participants.



Figure 7. **Gesture editing.** Top: We modify style from being neutral (left) to being sad (right) by combining the emotion latent from sad audio with the content latent from neutral audio. Bottom: We transform the style from Subject 13 being happy (left) to being angry (right) by merging the content latent from happy audio with the style and emotion latents from an angry audio of Subject 2.

tion methods [4, 55, 110] trained on coarse skeletal data.

5.3. Perceptual Study

Design. Our perceptual study is designed as a side-by-side comparison of two gesture videos generated with the same audio as input but by two different methods (AMUSE and another model or GT). The participants are asked to rate their preference of the methods on a five-point Likert scale for "synchronization with speech" and "gesture emotion appropriateness" given the GT emotion label of the input audio. We recruit 25 participants per method-to-method comparison on Amazon Mechnical Turk. Each participant is shown 24 pairs of randomly selected test set animations, 3 per emotion (neutral, happy, angry, sad, disgust, fear, surprise, and contempt). To allow the participant to get used to the task, we discard the answers of the first three comparisons and repeat these at the end. We incorporate three catch trials and responses from participants that fail on more than one are filtered out, as shown in Fig. 6 (right).

Results. The results of the study are shown in Fig. 6. AMUSE outperforms all competing methods by a considerable margin on both tasks, suggesting that AMUSE's generations are more appropriate for both the content of the input speech and its emotion compared to the baselines. However, it must be noted that there is still a significant gap between AMUSE and the GT. Please refer to the Sup. Mat. for details about the perceptual study.

5.4. Discussion and Future Work

Upper-body motion. We focus on the smooth coordination between the pelvis and upper body animation for side-by-side comparisons with other methods, as all other methods primarily focus on upper body movements. Future work should include lower-body motion and locomotion as these impact the perceived emotional state of a sequence.

Semantics. While the generated gestures, synchronized with the driving speech sequence, do not account for semantics such as deictic and metaphoric gestures, incorporating the text/language modality could help further improve in this direction.

Facial expressions. While emotional speech-driven face animation methods [18, 74] can be combined with bodies generated from AMUSE, jointly learning to generate emotional 3D bodies from speech is a topic that needs attention.

End-to-end training. Joint audio-gesture training may enhance results but requires careful loss term balancing and increased GPU memory. Therefore, we opted for separate training.

6. Conclusion

We present AMUSE, a framework to generate emotional body gestures from speech. The emotions and personal styles of the synthesized gestures can be controlled, thanks to the disentanglement of content, emotion, and style directly from the speech. The latent diffusion-based framework can further generate variations of the same gesture with the same emotion. Our quantitative evaluations show that AMUSE achieves state of the art performance on a variety of metrics: diversity, gesture emotion classification accuracy, Frechét gesture distance, beat alignment score, and semantic relevant gesture recall. Finally, our perceptual study demonstrates that AMUSE generates motions that are better synchronized and better match the emotion expressed of the input speech than previous state of the art.

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Disclosure. https://download.is.tue.mpg.de/amuse/Disclosure.txt

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